
title: D · Value-Added Components description: Eight strategic differentiators offered at no additional cost — bilingual UI, offline-first PWA, HR Copilot, payroll anomaly detector, hash-chained audit ledger, transparency portal, 10-year escrow, and a teacher-to-school placement optimiser.

D · Value-Added Components

!!! info “About this paper” Papers **A**, **B** and **C** describe the base scope demanded by the PBD. **This paper is different** — it describes eight **discretionary components** offered *in addition* to the base scope, at **no incremental cost** to the ABC of PHP 500 M. Each one is either a re-use of a component already scoped, a hardening of a mandatory requirement, or a discharge of a risk that would otherwise fall on DepEd. Formal contract wording is in [Paper B §16.5](#).

Design principle

Every value-added component (VAC) had to pass **three gates** before being included:

1. **Documented pain point.** The PBD acknowledges the need (or the operating context makes it obvious) but does not fully price a solution.
2. **Fits inside the 365 days.** No displacement of base scope; delivery slotted into an existing milestone.
3. **Survives audit.** Passes CSC, COA, NPC, and DICT review with no additional legal or procurement action.

Ideas that did not clear all three gates are recorded but not offered — because a VAC that becomes a scope conversation is a liability, not a differentiator.

The eight components

1 · Bilingual UI + SMS / USSD channels

What. Full user interface in Filipino, Cebuano, and Ilocano in addition to English. Payslip enquiries, leave applications, and OTP verification also reachable via SMS and USSD short-codes, so any teacher with a feature phone can transact.

Why. Roughly 60% of DepEd’s non-urban workforce still relies on SMS as a primary channel. Competitors typically ship English-only web/mobile.

Delivery. i18n framework at M2, translations delivered at M4, SMS/USSD gateway integrated at M4 alongside the notification service.

Base-scope tie-in. Extends the web and mobile UI scoped in §5 Architecture; the notification service already exists.

Evidence. Language coverage per module, SMS/USSD test-plan and gateway-provider certification (Annex V-01).

2 · Offline-first PWA with CRDT sync

What. Progressive Web App usable in schools with intermittent connectivity. Writes are captured locally using conflict-free replicated data types (CRDTs) and reconciled deterministically when the school reconnects. Kiosk daemon does the same for shared-device stations.

Why. The PBD requires “offline access” but does not specify the reconciliation strategy. Naive last-write-wins loses data at scale. CRDTs are proven — the same maths behind Google Docs and Apple Notes.

Delivery. PWA shell at M4; CRDT sync layer at M5; field-tested at three low-connectivity SDOs before M6 payroll gate.

Base-scope tie-in. Uses the same auth, encryption and schema as the online UI; only the transport and merge strategy differ.

Evidence. Reconciliation test-plan, chaos-test report showing zero-loss under 72-hour partition (Annex V-02).

3 · HR Copilot — self-hosted LLM

What. A conversational assistant grounded in the 201 file store, plantilla, DepEd Orders, and CSC memoranda. Answers natural-language questions such as *“How many secondary teachers in Region VIII are eligible for reclassification this fiscal year?”* Returns cited passages with the answer, never hallucinated free-text.

Why. Turns the HRIS from a database into a decision tool. The evaluation panel will visibly recognise this as a modern capability few competitors will demo credibly.

Delivery. Retrieval index at M5; assistant UI + guardrails at M7. Runs on a **self-hosted, open-weights model** (Llama-3.1-8B-Instruct or Qwen-2.5-7B-Instruct class) inside the DepEd DMZ — no data leaves the perimeter, satisfying RA 10173 by construction.

Base-scope tie-in. Uses the search/retrieval engine already scoped for document management and reporting.

Evidence. Grounding evaluation set with citation-accuracy scores, DPO opinion on data-flow topology, red-team results (Annex V-03).

4 · Payroll anomaly detector — M6 parallel-run gate

What. Statistical + rules-based anomaly detector that runs against every payroll cycle **before** disbursement is authorised. Flags: ghost employees, duplicate bank accounts, out-of-band net-pay deltas, deduction imbalances, GSIS/BIR/PhilHealth remittance drift, and time-clock anomalies.

Why. Turns the mandatory M6 parallel-run from a compliance chore into a **fraud-prevention asset**. Directly reduces DepEd’s post-audit exposure and the personal risk of the disbursing officer.

Delivery. Detector at M4 (training on legacy payroll data); gated integration at M6 alongside the parallel-run sign-off; deployed in production at M8.

Base-scope tie-in. Analytical extension of the payroll engine scoped in Paper C. Rules and thresholds are configurable by DepEd finance staff.

Evidence. Detection precision/recall on historical DepEd payroll (with anonymisation), operator runbook, escalation matrix (Annex V-04).

5 · Immutable, hash-chained audit ledger

What. Every state transition (personnel event, approval, payroll adjustment, permission change) is written to an append-only audit table whose rows are cryptographically hashed into a chain. A daily Merkle root is published to a public timestamp service (e.g. opentimestamps.org anchored to Bitcoin block headers).

Why. Blockchain-grade tamper-evidence at PostgreSQL cost. Any COA or Ombudsman investigator can independently verify that a given audit record existed on a given date and has not been altered.

Delivery. Schema at M2; hash-chain writer at M5; public timestamp job at M5; verification CLI shipped in the operations toolkit.

Base-scope tie-in. Extends the audit-log requirement in §7 Security. No new database, no new dependency — just a small extension of the existing audit schema plus a nightly cron.

Evidence. Verification demo, cryptographic protocol document, NPC-approved public-anchor plan (Annex V-05).

6 · Public transparency portal

What. A read-only, aggregate, anonymised public site: teacher-to-student ratios per region, plantilla vacancies by division, gender/age pyramid, average time-to-fill for CSC-mandated posts. No personally identifiable information ever leaves the perimeter — the portal is generated from a nightly snapshot with k -anonymity ≥ 5 applied to every cell.

Why. DepEd looks transparent, media get a self-serve source, civil-society researchers stop filing FOI requests for the same aggregates. Zero PII risk under RA 10173.

Delivery. Snapshot pipeline at M6; portal UI at M7; NPC-approved DPIA at M6.

Base-scope tie-in. Reuses the reporting engine, the same charts library, and the existing static-site build. No new authentication or infrastructure.

Evidence. DPIA report, k -anonymity certification, sample datasets (Annex V-06).

7 · 10-year source escrow + community edition

What. Two complementary continuity commitments beyond what the PBD minimum requires:

- **10-year source escrow** with a Philippine escrow agent — DepEd receives release triggers on bidder insolvency, missed critical patches, or agreed non-performance conditions.
- **Community edition** — a stripped-down variant of the HRIS core released under a permissive open-source licence, so other national agencies and LGUs can fork, contribute, or seed their own HRIS efforts.

Why. Directly answers the perennial procurement worry: *“What happens if the vendor disappears?”* Also positions DepEd as a leader in the government-tech open-source movement (aligning with DICT and DBM-PS policy direction).

Delivery. Escrow agreement executed at M1; community edition repository seeded at M8; ongoing quarterly public releases.

Base-scope tie-in. Extends the §11 Turnover commitments. No additional engineering cost beyond community-edition curation.

Evidence. Executed escrow deed, licence text, governance charter for the community project (Annex V-07).

8 · Teacher-to-school placement optimiser (advisory)

What. A recommendation engine that reads plantilla vacancies and current teacher assignments, and emits candidate reassignments with a trade-off analysis — for example, *“reassigning 12 teachers closes 8 subject-specific vacancies at the cost of a 47 km average commute increase.”* Every recommendation is reviewed by the appropriate authority (Schools Division Superintendent for intra-division moves; DepEd Central for cross-region moves). **No automated writes to any personnel record.** Every decision — accept, reject, or override — is logged to the audit ledger (VAC #5) with the human decision-maker, timestamp, and reason.

Why. Solves the perennial DepEd headline *“no math teachers in Mindanao”* by turning the plantilla module from a record store into a management decision tool. Import of a **proven-in-production pattern from Chile** (SIGE, ~240 K teachers) — not novel research. No competitor for this bid is likely to offer this.

Delivery. Data model in place at M4 (reads existing plantilla and 201 file tables). First solver runs at M6 in shadow mode against historical assignments. UI and human-in-the-loop decision workflow at M7. Production hand-off at M8.

Base-scope tie-in. Reads from the plantilla and 201 file tables defined in Paper C’s Core-HR and Recruitment bounded contexts. Writes only to a new advisory table plus the audit ledger. Reuses the workflow engine and approval routing that the base scope already ships.

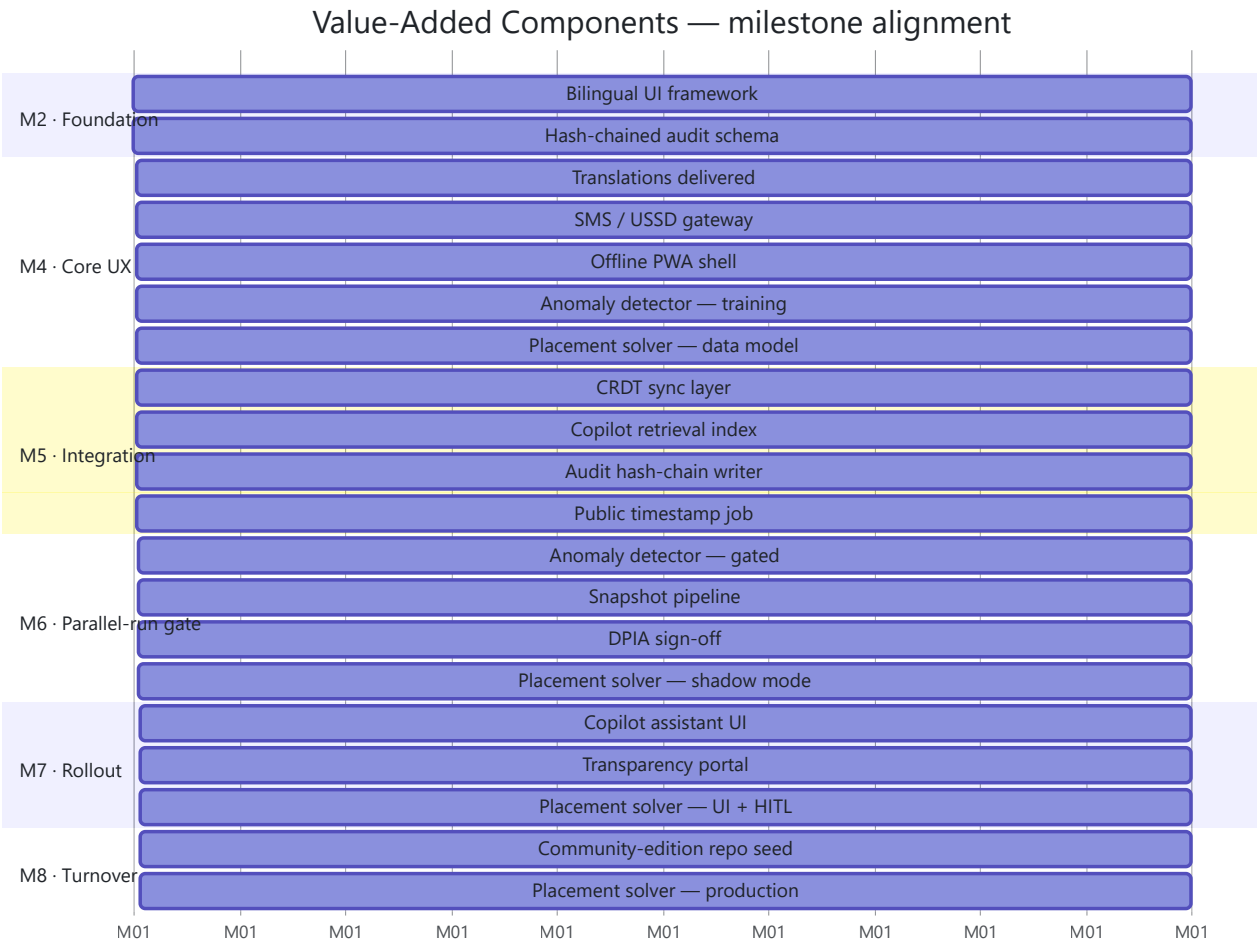
Political & governance guardrails.

- **Advisory only.** All final decisions remain with the CSC-authorized appointing authority. The solver never triggers a personnel record change.
- **Union-safe framing.** Every recommendation is accompanied by a natural-language rationale (subject match, distance, seniority, hardship-post premium) that the decision-maker can share with the affected teacher.
- **CSC rule engine.** Solver constraints include CSC merit-and-fitness rules and DepEd Orders on assignment; a recommendation that violates any hard rule is filtered before it reaches the UI.
- **DPO sign-off.** Data-flow is inside the perimeter (RA 10173 safe); DPIA delivered alongside the anomaly detector at M6.

Evidence. Solver technical design, CSC rule-set encoding, DPIA, shadow-mode evaluation report showing agreement rate with historical human decisions (Annex V-08).

Delivery timeline

The eight VACs are folded into the base milestone plan — no additional milestones, no critical-path additions:



Fit against the compare table

VAC	Reinforces paper
Bilingual UI + SMS/USSD	A (scope) · C (UI/notification service)
Offline PWA + CRDT	A (offline) · C (Container view)
HR Copilot	C (retrieval component) · B (SLA)
Payroll anomaly detector	A (M6 gate) · B (milestone entry/exit)
Hash-chained audit ledger	A (audit) · B (§7 Security response)
Public transparency portal	A (28 reports) · C (reporting engine)
10-year escrow + community edition	A (turnover) · B (§11 Continuity)
Placement optimiser (advisory)	C (Plantilla/Recruitment) · E (Chile SIGE precedent)

Formal offer wording

The formal, contractual wording of these VACs — including the “goodwill reduction” clause and delivery gating — is in [Paper B §16.5](#).

Related: [A](#) · [Brief](#) · [B](#) · [Response](#) · [C](#) · [Architecture](#) · [Downloads](#)